

Serie 20

DIAGONALIZABILITY, CAYLEY-HAMILTON

1. Let K be a field, let $n \geq 2$, and let

$$A = \begin{pmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ 0 & 0 & \cdots & 0 & -c_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{pmatrix} \in M_{n \times n}(K).$$

Prove that

$$\text{char}_A(X) = (-1)^n(X^n + c_{n-1}X^{n-1} + \cdots + c_0).$$

Hint: Use induction.

2. Let A be a $n \times n$ -matrix of rank r . Show that the degree of the minimal polynomial of A is smaller or equal than $r + 1$.
3. Prove that every 2×2 invertible real matrix belongs to one of the following categories:
- It is diagonalizable;
 - it is trigonalizable with algebraic multiplicity 2 and geometric multiplicity 1;
 - one can find a basis such that the matrix representation in that basis is

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \quad \text{with } b \neq 0.$$

4. Let K be a field, $A \in M_{n \times n}(K)$, and $p \in K[X]$ be a non-trivial polynomial such that $p(A) = 0$. Show that every eigenvalue of A is a root of p .

Hint: For an eigenvector v of A and a polynomial q over K , state and prove the relationship between v and $q(A) \cdot v$.

5. (a) Let A be a $n \times n$ -matrix. Prove that the subspace $\langle I_n, A, A^2, \dots \rangle$ of $M_{n \times n}(K)$ has dimension $\leq n$.

(b) Let $A := \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{pmatrix}$. Find a polynomial $p(X)$ with $p(A) = A^{-1}$.

6. Prove or disprove: There exists a real $n \times n$ -matrix A satisfying

$$A^2 + 2A + 5I_n = 0$$

if and only if n is even.

7. (a) Let f be an endomorphism of a finite-dimensional vector space V , and let $V = V_1 \oplus \dots \oplus V_r$ with f -invariant subspaces V_i . Show, that the arithmetic, resp. geometric multiplicities of an eigenvalue $\lambda \in K$ of f is equal to the sum of the arithmetic, resp. geometric multiplies of λ as an eigenvalue of the endomorphisms $f|_{V_i}$ of V_i .
- (b) Deduce that f is diagonalizable if and only if $f|_{V_i}$ is diagonalizable for every i .
- (c) Let f and g be endomorphisms for the same finite dimensional vector space V . Show that f and g are *simultaneously diagonalizable* (meaning that there exists a basis of eigenvectors of f which are all also eigenvectors of g) if and only if they commute and are diagonalizable.

Hint: To prove the the backward direction, first show that each eigenspace of f is g -invariant, i.e. that g maps eigenvectors of f to eigenvectors of f in the same eigenspace.

8. Let A, B be $n \times n$ matrices over a field F . Prove:

- (a) Prove that if $(I_n - AB)$ is invertible, then $(I_n - BA)$ is invertible, and

$$(I_n - BA)^{-1} = I_n + B(I_n - AB)^{-1}A$$

- (b) Use part (8a) to show that AB and BA have precisely the same characteristic values in F .

9. Let V be the vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$ which are continuous, i.e. the space of continuous real-valued functions on the real line. Let T be the linear operator on V defined by

$$(Tf)(x) = \int_0^x f(t)dt$$

Prove that T has no characteristic values.